AWS Cost Optimization

Are you using the most cost-effective techniques for your workloads? To make sure that you are, we will explore some popular cost optimisation techniques and good practices. Most of the services in AWS are instance based, so they are charged by the number of hours that the instance is utilised. A single instance hour refers to a regular clock hour for which an instance was available to you, regardless of usage. AWS' variety of services and pricing options offers a flexible way to manage your resources in an efficient way and still meet your business needs. These are some of the commonly used practices and techniques to optimise the cost for AWS services :

1. Proper Sizing of Instances

AWS offers more than 60 EC2 instance types. Going for a very large instance type with higher compute is not always a prudent choice. Properly sizing the instances means choosing the cheapest instance but still meeting the performance requirements of the application. Apart from this, you should monitor your production workflow frequently and if the instances are underutilised (i.e. not using the full compute capacity), you can downsize them to a lower instance type which can definitely save a lot of money for you.

Here is a very good example of over-provisioning the EC2 instances. Suppose you have started up and you are deploying your product on AWS. Your application can run on an m4.large (which costs 0.1 $ per hour) instance but you haven't given much thought about right sizing and you don't want to compromise on performance, so, you go for m4.xlarge instance type (which costs $0.2 per hour). If you run this instance for an year you spend $1752 but if you would have provisioned it right at the beginning you would have paid half that amount i.e. $876. So, just by going one size up, you are literally paying double the money.

There is an EC2 Right sizing solution created by AWS, which is a template and can be launched using Cloudformation. It runs your application on different instance types and creates a detailed report of usage, compute and I/O, at the end, which can used to determine the right size of instance for your needs.

2. Buying Reserved Instances

If you are planning to use AWS for long term, purchasing a reserved capacity instead of using On Demand instances is always a wise choice. AWS offers a significant discount of up to 25% off on the price of On